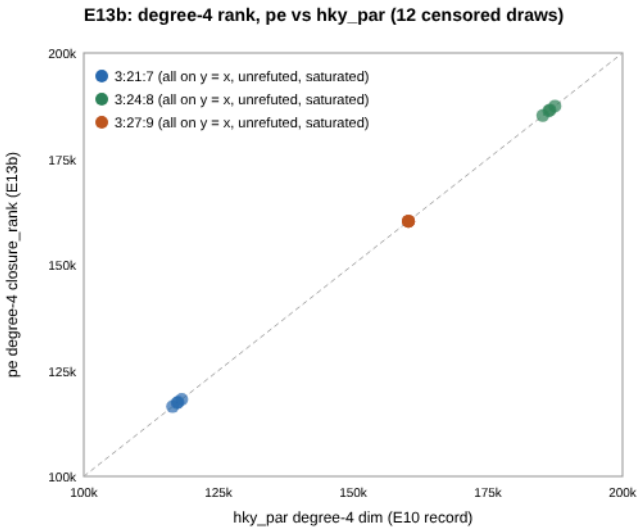
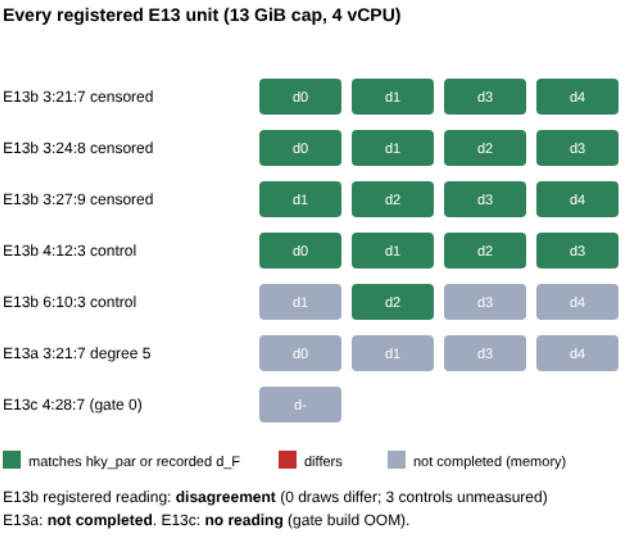
E13: independent-engine check of the E10 $d_F \geq 5$ draws, degree 5 at (3,21,7), and (4,28,7)

Study `research/fall_degree_bounds_20261008`, Amendments 17 (design, readings, predictions) and 18 (local route: 4 vCPU, 15 GB, `ulimit -v` 13 GiB, sequential). Source: frozen archive of main `a2be6609` (sha256 `18b14930…`). Run 2026-11-07 22:56 → 2026-11-08 10:05 UTC, one container restart (01:37, B_3-21-7_d0 recomputed from scratch, attempt 1 kept). Gates G0 (7 tests ok) and G1 (closure_rank 131,588, diff empty) passed. Scored by `e13/score_e13.py` from committed files → `e13/score.json`.





Registered readings

part	reading (rule as written)	recorded prediction
E13b	**disagreement** — no draw differs; 3 of 8 controls (6:10:3 d1, d3, d4, N = 58) not completed (memory), so “every control refutes” is not met	independent agreement 0.85 (A17); N=58 controls not completed 0.5 (A18)
E13a	**not completed** — 0 of 4 draws measured; each exceeded 13 GiB at degree 5 after ~19 min	d_F=5 0.6, ≥6 0.3, not completed 0.1 (A17); not completed (memory) 0.85 (A18)
E13c	**no reading** — gate 0 did not build: rustc/LLVM out of memory under the 13 GiB cap applied to the whole `run_window.sh`; no draw computed; consistency check with E12c r3 draw 0 not possible	no reading 0.8 (A17); not completed (memory) 0.95 (A18)

E13b censored draws (pe, --to 4, --max-cols 400000)

cell	draw	hky_par dim_4	pe $rank_4$	dim_3 hky / pe	status	s
3:21:7	0	117,335	117,335	2,338 / 2,338	agree	579
3:21:7	1	116,485	116,485	2,338 / 2,338	agree	511
3:21:7	3	118,185	118,185	2,338 / 2,338	agree	533
3:21:7	4	117,505	117,505	2,338 / 2,338	agree	586
3:24:8	0	186,549	186,549	3,056 / 3,056	agree	1850
3:24:8	1	187,453	187,453	3,056 / 3,056	agree	1827
3:24:8	2	186,323	186,323	3,056 / 3,056	agree	1812
3:24:8	3	185,193	185,193	3,056 / 3,056	agree	1813
3:27:9	1	160,259	160,259	3,870 / 3,870	agree	1924
3:27:9	2	160,259	160,259	3,870 / 3,870	agree	1903

3:27:9	3	160,259	160,259	3,870 / 3,870	agree	1944
3:27:9	4	160,259	160,259	3,870 / 3,870	agree	1934

Every censored draw: degree 4 saturated, not refuted, rank equal to hky_par's. The suspension clause of the “disagreement” reading lists no cell, because no draw differs.

E13b controls

cell	draw	recorded d_F	pe closure degree	status	s
4:12:3	0	4	4	refutes at recorded d_F	124
4:12:3	1	4	4	refutes at recorded d_F	133
4:12:3	2	4	4	refutes at recorded d_F	131
4:12:3	3	4	4	refutes at recorded d_F	142
6:10:3	1	4	—	not completed (memory)	2343
6:10:3	2	3	3	refutes at recorded d_F	13
6:10:3	3	4	—	not completed (memory)	2594
6:10:3	4	4	—	not completed (memory)	2424

E13a: degree-5 closure at (3,21,7)

draw	status	s
0	unmeasured (not completed, memory)	1084
1	unmeasured (not completed, memory)	1133
3	unmeasured (not completed, memory)	1139
4	unmeasured (not completed, memory)	1135

What this does and does not show

- Shows: a second closure implementation (partial enlargement, shares only `build_decomposition_system`) reproduces hky_par's degree-3 and degree-4 ranks on all 12 E10 $d_F \geq 5$ draws, and reproduces the recorded refutation on the 5 controls it could finish (4:12:3 ×4 at degree 4, 6:10:3 d2 at degree 3). This removes the “same-code only” objection for those draws.
- Does not show: the positive controls at N = 58 refuting at degree 4 under the second engine, so the engine's ability to detect a degree-4 refutation was confirmed only at N = 48 (4:12:3); the exact d_F at (3,21,7); anything at (4,28,7). These need more than 13 GiB.
- No improvement over the index-calculus baseline is claimed; this is a measurement of degree-of-regularity structure, not a solve. No leaderboard or benchmark row.